The laws of exponents and scientific notation

1 Use scientific notation to complete the expression.

$$1 \times 10^{-2} \times 2 \times 10^{-5} = 2 \times \square$$

2 Evaluate the following. Express your answer in scientific notation.

a $3 \times 10^5 \times 7 \times 10^2$

b $\left(4.3 \times 10^{13}\right) \times \left(1.3 \times 10^{-6}\right)$

c $\left(9.3 \times 10^{12}\right) \times \left(8.4 \times 10^8\right)$

d $\left(5.2 \times 10^{20}\right) \times \left(9.6 \times 10^{12}\right)$

e $\left(9 \times 10^9\right) \times \left(5.5 \times 10^{-6}\right)$

f $\left(41.83 \times 10^{28}\right) \div \left(4.7 \times 10^{18}\right)$

g $1.77 \times 10^9 + 3.24 \times 10^{10}$

h $2.35 \times 10^{-7} - 4.72 \times 10^{-8}$

i $\dfrac{73.8 \times 10^{17}}{9 \times 10^{12}}$

j $\dfrac{3 \times 10^3}{12 \times 10^{-1}}$

k $\dfrac{800\,000 \times 180\,000\,000}{6\,000\,000\,000}$

l $11\,000\,000 \times 0.004 \times 0.0005 \times 20\,000$

m $4.088 \times 10^{15} \div \left(5.6 \times 10^8\right)$

3 Evaluate $\left(3.7 \times 10^{-4}\right)^2$. Give your answer in scientific notation, correct to five significant figures

4 Evaluate $\dfrac{6.45 \times 10^9 \times 4.3 \times 10^{-3}}{8.3 \times 10^4}$. Give your answer in scientific notation, correct to four decimal places.

5 A country's land size is approximately $84\,000 \text{ mi}^2$ and its population is approximately 21 million people.

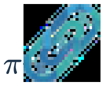
a Express the following in scientific notation:

i The land size in square miles

ii The population

b Determine the population density measured in $\dfrac{\text{mi}^2}{\text{person}}$ and write using scientific notation.

6 The speed of light is approximately $3 \times 10^5 \dfrac{\text{km}}{\text{s}}$. If a star that is 7.525×10^{12} km away from Earth is born, how long after it is born would we see it from Earth? Give your answer in scientific notation, rounding to 1 significant figure.

7 The volume of a sphere is given by $V = \frac{4}{3}\pi$ 
Planet A has a radius of 4×10^{10} km, while Planet B has a radius of 8×10^{17} km.
How many times greater than the volume of Planet A is the volume of Planet B? Give your answer in scientific notation.

8 The average mass of a grain of sand on a beach is about 1.5×10^{-5} g. Over several years, a beach goes through a natural process of weathering and loses approximately 9.5×10^{10} grains of sand.
What mass of sand is lost over this time period? Give your answer in scientific notation.

9 The mass of a planet is approximately 8.1×10^{21} kg and the mass of its single moon is 7.9×10^{19} kg. Determine the combined mass of the planet and its moon in scientific notation.

Additional questions

10 Evaluate each of the following expressions. Write your answers using scientific notation:

a $2 \times 10^4 + 6 \times 10^4$
b $9 \times 10^3 - 4 \times 10^3$
c $1 \times 10^{-15} + 7 \times 10^{-15}$
d $(4 \times 10^6) \times (3 \times 10^6)$
e $6 \times 10^{-9} - 5 \times 10^{-9}$
f $(3 \times 10^{-2}) \div (4 \times 10^{-2})$
g $3 \times 10^4 + 2 \times 10^5$
h $2 \times 10^3 - 4 \times 10^2$
i $(5 \times 10^{-15}) \div (2 \times 10^8)$
j $(4 \times 10^{-4}) \times (3 \times 10^{-9})$
k $1 \times 10^{11} - 8 \times 10^{10}$
l $(9 \times 10^2) \div (2 \times 10^1)$
m $3.5 \times 10^8 - 2.1 \times 10^6$
n $2.2 \times 10^9 - 1.5 \times 10^7$
o $(4.9 \times 10^5) \times (7.1 \times 10^8)$
p $(1.75 \times 10^6) \div (1.25 \times 10^8)$
q $(1.8 \times 10^8) \div (8 \times 10^2)$
r $(2.8 \times 10^{18}) \times (1.4 \times 10^{-20})$

11 Evaluate each of the following expressions, writing your answer in standard form:

a $3.9 \times 10^2 + 4 \times 10^3$
b $5.8 \times 10^{-3} + 1.3 \times 10^{-4}$
c $7.3 \times 10^{-5} - 2.5 \times 10^{-5}$
d $9.4 \times 10^{-6} - 4.3 \times 10^{-7}$

12 Evaluate each of the following expressions. [redacted] your answers using scientific notation:

a 0.45×30

b $220000 + 5700$

c $7500000000 \times 30000000$

d $0.0009 \div 0.004$

e $63200000 + 107000$

f $0.0000925 - 0.00009$

g $0.00083 + 0.0573$

h $356775000000 \div 7.1$

i $388.08 - 90.09$

j 405×5.88

k 5000.87×100000

l $20.41 \div 4000$